🟢 Easy (10 Questions)

1.

What is the core of the Android operating system?
A) Windows Kernel
B) Linux Kernel
C) macOS Kernel
D) Unix Kernel
Answer: B
Explanation: Android is built on top of the Linux Kernel, which provides low-level system functions.

2.

Which language is primarily used to write Android apps?
A) Python
B) Java/Kotlin
C) C#
D) Swift
Answer: B
Explanation: Android apps are mainly developed using Java or Kotlin.

3.

What is the file format of an Android app package?
A) .exe
B) .apk
C) .jar
D) .app
Answer: B
Explanation: Android apps are packaged as APK (Android Package) files.

4.

Which Android component represents a single screen in an app?
A) Service
B) Content Provider
C) Activity
D) Broadcast Receiver
Answer: C
Explanation: Activities provide the UI for a single screen or interaction.

5.

Where does Android store user app data?
A) /system
B) /cache
C) /data
D) /bin
Answer: C
Explanation: User and app data are stored under the /data directory.

6.

What is the purpose of Android Runtime (ART)?
A) To compile apps at runtime
B) To execute apps using compiled bytecode
C) To manage hardware drivers
D) To update the operating system
Answer: B
Explanation: ART executes compiled bytecode of Android apps efficiently.

7.

Which Android security feature isolates apps from each other?
A) Encryption
B) Sandboxing
C) VPN
D) Firewall
Answer: B
Explanation: Sandboxing runs each app in a separate process with unique user ID.

8.

Which Android build step converts Java bytecode into Dalvik bytecode?
A) Dexing
B) Packaging
C) Signing
D) Compiling
Answer: A
Explanation: Dexing converts compiled Java bytecode (.class files) into .dex files.

9.

What permission system does Android use to protect sensitive device features?
A) Access Control List (ACL)
B) User Roles
C) App Permissions
D) Firewall rules
Answer: C
Explanation: Apps must request permissions to access camera, contacts, location, etc.

10.

What does rooting an Android device enable?
A) Use device in airplane mode
B) Access to system-level files and commands
C) Install apps from Google Play
D) Increase screen brightness
Answer: B
Explanation: Rooting provides superuser access to modify system files.

🔸 Medium (20 Questions)

11.

Which directory contains the core Android operating system files?
A) /cache
B) /system
C) /data
D) /storage
Answer: B
Explanation: The /system directory holds essential OS files and system apps.

12.

Which Android app component is responsible for running background tasks without a UI?
A) Activity
B) Service
C) Broadcast Receiver
D) Content Provider
Answer: B
Explanation: Services run background tasks independent of the user interface.

13.

What role does the Android Hardware Abstraction Layer (HAL) play?
A) Connect apps directly to hardware
B) Translate hardware signals for the OS
C) Run apps in sandbox
D) Provide UI elements
Answer: B
Explanation: HAL acts as an interface between hardware and higher-level APIs.

14.

Which tool is used to package and sign an Android app?
A) Android Debug Bridge (ADB)
B) Android Studio Build System
C) Google Play Console
D) Gradle
Answer: B
Explanation: Android Studio build system compiles, packages, and signs APK files.

15.

What does the 'Broadcast Receiver' component do?
A) Display UI
B) Handle system-wide or app-wide broadcasts/events
C) Store app data
D) Manage network connections
Answer: B
Explanation: Broadcast receivers listen to and respond to system or app events.

16.

What is the main function of SELinux in Android?
A) Manage user interface
B) Enforce mandatory access control policies
C) Schedule tasks
D) Manage app installation
Answer: B
Explanation: SELinux enforces strict access control for security.

17.

Which process ensures the Android device boots with trusted software?
A) Verified Boot
B) Fast Boot
C) Safe Mode
D) Root Access
Answer: A
Explanation: Verified Boot checks software integrity during device startup.

18.

Which Android file system directory is typically mounted as external storage?
A) /system
B) /data
C) /sdcard or /storage
D) /cache
Answer: C
Explanation: External storage for media and documents is mounted under /sdcard or /storage.

19.

Which of the following is NOT a valid Android app component?
A) Activity
B) Fragment
C) Service
D) Broadcast Receiver
Answer: B
Explanation: Fragment is part of an activity, not a standalone component.

20.

Which Android security model component assigns unique Linux user IDs to apps?
A) Package Manager
B) Linux Kernel
C) Android Runtime
D) Dalvik VM
Answer: B
Explanation: The Linux Kernel provides unique user IDs isolating apps.

21.

What does ProGuard do during Android app building?
A) Run unit tests
B) Shrink, optimize, and obfuscate code
C) Manage app permissions
D) Sign APKs
Answer: B
Explanation: ProGuard reduces app size and makes code harder to reverse engineer.

22.

Why is app signing important in Android?
A) To compress the APK
B) To verify app source and integrity
C) To encrypt user data
D) To enable multitasking
Answer: B
Explanation: Signing ensures apps are from trusted developers and untampered.

23.

Which Android system service handles inter-app communication?
A) Content Provider
B) Service
C) Broadcast Receiver
D) Activity Manager
Answer: A
Explanation: Content Providers enable data sharing between apps.

24.

How does Android enforce app permissions at runtime?
A) Through user prompts during app installation
B) Through permission requests at app runtime
C) By blocking apps without permissions from launching
D) Permissions are not enforced
Answer: B
Explanation: Since Android 6.0, dangerous permissions require runtime user approval.

25.

Which tool is used to debug Android apps via command line?
A) Android Debug Bridge (ADB)
B) Gradle
C) ProGuard
D) Android Studio
Answer: A
Explanation: ADB allows developers to run commands and debug apps on devices.

26.

Which Android security mechanism restricts access to sensitive APIs?
A) VPN
B) Runtime Permissions

C) Firewall
D) Encryption
Answer: B
Explanation: Apps must request permission to access camera, contacts, etc., at runtime.

27.

What is the first step in the Android app build process?
A) Signing
B) Compilation of source code
C) Packaging into APK
D) Dexing
Answer: B
Explanation: Source code is first compiled into bytecode before other steps.

28.

Which statement about rooting is true?
A) Rooting is officially supported by Google
B) Rooting allows full control but can void warranty and reduce security
C) Rooting improves battery life significantly
D) Rooting disables app installation
Answer: B
Explanation: Rooting gives superuser access but voids warranty and increases risk.

29.

Which directory stores temporary system files?
A) /system
B) /cache
C) /data
D) /proc
Answer: B
Explanation: /cache stores temporary files for system use.

30.

What is the purpose of 'Verified Boot' in Android?
A) To speed up boot time
B) To ensure device boots with untampered software
C) To allow multiple users
D) To install updates automatically
Answer: B
Explanation: Verified Boot prevents booting if system integrity is compromised.

Hard (10 Questions)

31.

How does the Android Runtime (ART) differ from Dalvik?
A) ART executes bytecode at runtime, Dalvik precompiles
B) ART precompiles bytecode ahead of time, Dalvik uses Just-in-Time compilation
C) ART is deprecated, Dalvik is the latest runtime
D) ART only runs on emulators
Answer: B
Explanation: ART compiles apps ahead of time for better performance, Dalvik used JIT compilation.

32.

Which security model enforces mandatory access control in Android?
A) DAC (Discretionary Access Control)
B) SELinux
C) ACL (Access Control List)
D) RBAC (Role-Based Access Control)
Answer: B
Explanation: SELinux enforces mandatory access control policies in Android.

33.

Which Android system layer abstracts hardware details from the OS?
A) Application Framework
B) Android Runtime
C) Hardware Abstraction Layer (HAL)
D) Linux Kernel
Answer: C
Explanation: HAL provides standard interfaces to hardware, hiding specifics from the OS.

34.

What risks does rooting an Android device introduce?
A) Increased security and system updates
B) Potential malware installation and data theft
C) Enhanced battery management
D) None; rooting only improves device
Answer: B
Explanation: Rooting can expose the device to malicious apps with elevated privileges.

35.

Which Android build tool manages dependencies and automates builds?
A) Maven
B) Gradle
C) Ant
D) Jenkins
Answer: B
Explanation: Gradle is the official Android build automation tool.

36.

What is the significance of the ‘android:exported’ attribute in app components?
A) Determines if a component can be accessed by other apps

B) Controls app permissions
C) Specifies app version
D) Enables device rooting
Answer: A
Explanation: This attribute controls component exposure to other apps.

37.

Which Android security feature restricts inter-app data sharing by default?
A) Permissions
B) Sandboxing
C) Verified Boot
D) Encryption
Answer: B
Explanation: Sandboxing isolates app data to prevent unauthorized access.

38.

Which of the following is NOT a common Android app vulnerability?
A) SQL Injection
B) Buffer Overflow
C) Cross-site scripting
D) Infinite loop errors
Answer: D
Explanation: Infinite loops cause performance issues but are not a security vulnerability.

39.

How does ProGuard help secure Android apps?
A) Encrypts user data
B) Obfuscates code to prevent reverse engineering
C) Blocks unauthorized app installs
D) Manages app permissions
Answer: B
Explanation: ProGuard renames classes/methods making code analysis harder for attackers.

40.

What does a 'Signature-based' app verification check during app installation?
A) App size
B) Developer's digital signature to verify source and integrity
C) User permissions
D) Runtime errors
Answer: B
Explanation: Signature verification ensures the app is from a trusted developer and untampered.